

IR-7 Additive Nomenclature

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IR-7.1 INTRODUCTION

IR-7.1.1 General

Additive nomenclature was originally developed for Werner-type coordination compounds, which were regarded as composed of a central atom (or atoms) surrounded by added groups known as ligands, but many other types of compound may also be conveniently given additive names. Such names are constructed by placing the names of the ligands (sometimes modified) as prefixes to the name(s) of the central atom(s).

This Chapter deals with the general characteristics of additive nomenclature and provides examples of additive names for simple mononuclear and polynuclear compounds. Chain and ring compounds are then treated using additive principles supplemented by further conventions. Additive names for inorganic acids are discussed in Chapter IR-8. Additive nomenclature as applied to metal coordination compounds is described in further detail in Chapter IR-9 (where a flowchart, Figure IR-9.1, provides a general procedure for naming coordination compounds). Additive names for a large number of simple compounds are given in Table IX*.

* Tables numbered with a Roman numeral are collected together at the end of this book.

Note that in some cases, a compound named additively may alternatively and equally systematically be named substitutively on the basis of a suitably chosen parent structure (Chapter IR-6). It is important to note, however, that additive names for parent hydrides cannot be used as parent names in substitutive nomenclature.

IR-7.1.2 **Choosing a central atom or atoms, or a chain or ring structure**

Making a choice of central atom or atoms is a key step in the process of naming a compound using additive nomenclature. If there are (one or more) metal atoms in the compound, these should be chosen as the central atom(s). Such atom(s) should also be relatively central in the structure and, where possible, should be chosen to make use of molecular symmetry (thereby shortening the name). Usually hydrogen atoms are disregarded when choosing central atoms.

For some compounds, a choice of central atom or atoms will remain. The atom(s) that occur(s) latest when following the arrow in Table VI should be chosen as the central atom(s).

If there is more than one central atom in a structure according to the above criteria then the compound can be named as a dinuclear or polynuclear compound.

As an alternative to the procedure above, a group of atoms forming a chain or ring sub-structure within a compound may be chosen in order to give the compound an additive name using the chains and rings nomenclature outlined in Section IR-7.4.

IR-7.1.3 **Representing ligands in additive names**

Additive names are constructed by placing (sometimes modified) ligand names as prefixes to the name of the central atom. For anionic ligands, the anion endings 'ide', 'ate' and 'ite' (see Section IR-5.3.3) are changed to 'ido', 'ato' and 'ito', respectively, when generating these prefixes. Names of neutral and cationic ligands are used unchanged, except in a few special cases, most notably water (prefix 'aqua'), ammonia (prefix 'ammine'), carbon monoxide bound through carbon (prefix 'carbonyl'), and nitrogen monoxide bound through nitrogen (prefix 'nitrosyl') (*cf.* Section IR-9.2.4.1).

In principle, it is a matter of convention whether a ligand is considered to be anionic, neutral or cationic. The default is to consider ligands as anionic, so that OH is 'hydroxido', Cl 'chlorido', SO₄ 'sulfato', *etc.* Some ligands are conventionally regarded as neutral, *e.g.* amines and phosphanes and ligands derived from hydrocarbons by removal of a hydrogen atom, such as methyl, benzyl, *etc.*

Appropriate prefixes to represent many simple ligands within names are given in Table IX. For further details, see Section IR-9.2.2.3.

IR-7.1.4 **Ions and radicals**

Anionic species take the ending 'ate' in additive nomenclature, whereas no distinguishing termination is used for cationic or neutral species. Additive names of ions end with the charge number (see Section IR-5.4.2.2). In additive names of radicals, the radical character of the compound may be indicated by a radical dot, •, added in parentheses and centred, after the name of the compound. Polyradicals are indicated by the appropriate numeral placed before the dot. For example, a diradical is indicated by '(2•)'.

IR-7.2

MONONUCLEAR ENTITIES

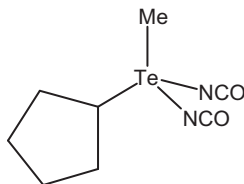
Names of mononuclear compounds and ions, *i.e.* of species with a single central atom, are formed by citing the appropriate prefixes for the ligands alphabetically before the name of the central atom. Ligands occurring more than once are collected in the name by means of multiplicative prefixes (Table IV), *i.e.* ‘di’, ‘tri’, ‘tetra’, *etc.*, for simple ligands such as chlorido, benzyl, aqua, ammine and hydroxido, and ‘bis’, ‘tris’, ‘tetrakis’, *etc.*, for more complex ligands, *e.g.* 2,3,4,5,6-pentachlorobenzyl and triphenylphosphane. The latter prefixes are also used to avoid any ambiguity which might attend the use of ‘di’, ‘tri’, *etc.* Multiplicative prefixes which are not inherent parts of the ligand name do not affect the alphabetical ordering.

Prefixes representing ligands can be separated using enclosing marks (see also Section IR-9.2.2.3), and this should be done for all but the simplest ligands, including organic ligands. In some cases the use of enclosing marks is essential in order to avoid ambiguity, as in Examples 10 and 11 below.

In several of the examples below, substitutive names (see Chapter IR-6) are also given. In some cases, however, there is no parent hydride available for the construction of a substitutive name (see Examples 9 and 11). Note also that the formulae given below in square brackets are coordination compound-type formulae with the central atom listed first.

Examples:

- | | |
|---|---|
| 1. $\text{Si}(\text{OH})_4$ | tetrahydroxidosilicon (additive),
or silanetetrol (substitutive) |
| 2. $\text{B}(\text{OMe})_3$ | trimethoxidoboron or
tris(methanolato)boron (both additive),
or trimethoxyborane (substitutive) |
| 3. FCIO or $[\text{ClFO}]$ | fluoridooxidochlorine (additive),
or fluoro- λ^3 -chloranone (substitutive) |
| 4. ClOCl or $[\text{OCl}_2]$ | dichloridooxygen (additive),
or dichlorooxidane (substitutive) |
| 5. $[\text{Ga}\{\text{OS}(\text{O})\text{Me}\}_3]$ | tris(methanesulfinato)gallium (additive),
or tris(methanesulfinyloxy)gallane (substitutive) |
| 6. $\text{MeP}(\text{H})\text{SiH}_3$ or
$[\text{SiH}_3\{\text{P}(\text{H})\text{Me}\}]$ | trihydrido(methylphosphanido)silicon (additive),
or methyl(silyl)phosphane (substitutive) |
| 7. $\text{NH}^{2\bullet}$ | hydridonitrogen($2\bullet$) (additive),
or azanylidene (substitutive) |
| 8. $\text{HOC}(\text{O})^\bullet$ | hydroxidooxidocarbon($\bullet$) (additive),
or hydroxyoxomethyl (substitutive) |
| 9. FArH or $[\text{ArFH}]$ | fluoridohydridoargon |
| 10. $[\text{HgMePh}]$ | methyl(phenyl)mercury (additive) |
| 11. $[\text{Hg}(\text{CHCl}_2)\text{Ph}]$ | (dichloromethyl)(phenyl)mercury |

12. $[\text{Te}(\text{C}_5\text{H}_9)\text{Me}(\text{NCO})_2]$ 

bis(cyanato-*N*)(cyclopentyl)(methyl)tellurium (additive),
or cyclopentyldiisocyanato(methyl)- λ^4 -tellane (substitutive)

- | | |
|---|---|
| 13. $[\text{Al}(\text{POCl}_3)_6]^{3+}$ | hexakis(trichloridooxidophosphorus)aluminium(3+) |
| 14. $[\text{Al}(\text{OH}_2)_6]^{3+}$ | hexaaquaaluminium(3+) |
| 15. $[\text{H}(\text{py})_2]^+$ | bis(pyridine)hydrogen(1+) |
| 16. $[\text{H}(\text{OH}_2)_2]^+$ | diaquahydrogen(1+) |
| 17. $[\text{BH}_2(\text{py})_2]^+$ | dihydridobis(pyridine)boron(1+) |
| 18. $[\text{PFO}_3]^{2-}$ | fluoridotrioxidophosphate(2-) |
| 19. $[\text{Sb}(\text{OH})_6]^-$ | hexahydroxidoantimonate(1-) (additive),
or hexahydroxy- λ^5 -stibanuide (substitutive) |
| 20. $[\text{HF}_2]^-$ | difluoridohydrogenate(1-) |
| 21. $[\text{BH}_2\text{Cl}_2]^-$ | dichloridodihydridoborate(1-) (additive),
or dichloroboranuide (substitutive) |
| 22. $\text{OCO}^{\bullet-}$ | dioxidocarbonate($\bullet 1-$) |
| 23. $\text{NO}^{(2\bullet)-}$ | oxidonitrate($2\bullet 1-$) |
| 24. $\text{PO}_3^{\bullet 2-}$ | trioxidophosphate($\bullet 2-$) |
| 25. $[\text{ICl}_2]^+$ | dichloridoiodine(1+) (additive),
or dichloroiodanium (substitutive) |
| 26. $[\text{BH}_4]^-$ | tetrahydridoborate(1-) (additive),
or boranuide (substitutive) |
| 27. CH_5^- | pentahydridocarbonate(1-) (additive),
or methanuide (substitutive) |
| 28. $[\text{PH}_6]^-$ | hexahydridophosphate(1-) (additive),
or λ^5 -phosphanuide (substitutive) |
| 29. $[\text{PF}_6]^-$ | hexafluoridophosphate(1-) (additive),
or hexafluoro- λ^5 -phosphanuide (substitutive) |

IR-7.3 POLYNUCLEAR ENTITIES

IR-7.3.1 Symmetrical dinuclear entities

In symmetrical dinuclear entities, each of the central atoms is of the same element and they are identically ligated. Below, additive names of several formats are given to a number of

such species. Again, in some cases substitutive names are also easily constructed, as exemplified below.

The general procedure for naming a symmetrical dinuclear entity is as follows.

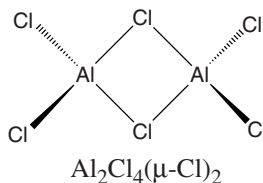
The ligands are represented in the usual way and the multiplicative affix 'di' is added immediately before the name of the central atom. The name of the central element is modified to the 'ate' form if the compound is an anion.

A bond between the two central atoms, if there is one, is indicated by adding to the name the italicized symbols for those two atoms, separated by an 'em' dash and enclosed in parentheses.

In bridged dinuclear species, bridging ligands are indicated by the Greek letter μ , placed before the ligand name and separated from it by a hyphen. The whole term, *e.g.* ' μ -chlorido', is separated from the rest of the name by hyphens. If the bridging ligand occurs more than once, multiplicative prefixes are employed (see also Sections IR-9.1.2.10 and IR-9.2.5.2).

Examples:

1. $[\text{Et}_3\text{PbPbEt}_3]$ hexaethyllead(*Pb—Pb*) (additive),
or hexaethyldiplumbane (substitutive)
2. $\text{HSSH}^{\bullet-}$ dihydridodisulfate(*S—S*)($\bullet 1-$) (additive),
or disulfanuidyl (substitutive)
3. NCCN dinitridodicarbon(*C—C*)
4. $\text{NCCN}^{\bullet-}$ dinitridodicarbonate(*C—C*)($\bullet 1-$)
5. $(\text{NC})\text{SS}(\text{CN})$ bis(nitridocarbonato)disulfur(*S—S*),
or dicyanidodisulfur(*S—S*)
6. $(\text{NC})\text{SS}(\text{CN})^{\bullet-}$ bis(nitridocarbonato)disulfate(*S—S*)($\bullet 1-$),
or dicyanidodisulfate(*S—S*)($\bullet 1-$)
7. OCIO μ -chlorido-dioxygen,
or dioxidochlorine
- 8.



di- μ -chlorido-tetrachloridodialuminium

A variant of the format in the above additive names involves starting with 'bis' and then citing the name of the half-molecule or ion in parentheses. Thus, Examples 1–6 and 8 become:

Examples:

9. $[\text{Et}_3\text{PbPbEt}_3]$ bis(triethyllead)(*Pb—Pb*)
10. $\text{HSSH}^{\bullet-}$ bis(hydridosulfate)(*S—S*)($\bullet 1-$)
11. NCCN bis(nitridocarbon)(*C—C*)
12. $\text{NCCN}^{\bullet-}$ bis(nitridocarbonate)(*C—C*)($\bullet 1-$)

- | | |
|---|---|
| 13. (NC)SS(CN) | bis[(nitridocarbonato)sulfur](S—S),
or bis(cyanidosulfur)(S—S) |
| 14. (NC)SS(CN) ^{•-} | bis[(nitridocarbonato)sulfate](S—S)(•1—),
or bis(cyanidosulfate)(S—S)(•1—) |
| 15. Cl ₂ Al(μ-Cl) ₂ AlCl ₂ | di-μ-chlorido-bis(dichloridoaluminium) |

Note that the five compounds in Examples 10–14 may also easily be named as chain compounds, as shown in Section IR-7.4. The name in Example 14 differs from that given in Ref. 1 (in which the sulfur–sulfur bond was indicated as above, but the carbon atoms were treated as the central atoms).

The species in Examples 13 and 14 may also be regarded as containing a bridging ligand, as demonstrated in Examples 16 and 17.

Examples:

- | | |
|----------------------------|---|
| 16. [NCSSCN] | μ-disulfanediido-bis(nitridocarbon) |
| 17. [NCSSCN] ^{•-} | μ-disulfanediido-bis(nitridocarbonate)(•1—) |

IR-7.3.2

Non-symmetrical dinuclear compounds

There are two types of non-symmetrical dinuclear compounds: (i) those with identical central atoms differently ligated, and (ii) those with different central atoms. In both cases names are formed by means of the procedure described in Section IR-9.2.5, which also deals with bridging groups.

Priority is assigned to the central atoms as follows. For cases of type (i) the central atom carrying the greater number of alphabetically preferred ligands is numbered 1. For cases of type (ii) the number 1 is assigned to the higher priority central element of Table VI, whatever the ligand distribution.

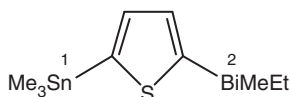
In both types of compound, names are constructed in the usual way, by first citing the prefixes representing the ligands in alphabetical order. Each prefix representing a ligand is followed by a hyphen, the number(s) assigned to the central atom(s) to which the ligand is attached (see below), the Greek letter κ (kappa) (see Section IR-9.2.4.2) with a right superscript denoting the number of such ligands bound to the central atom(s) (the number 1 being omitted for a single ligand), and the italic element symbol for the ligating atom by which the ligand is attached to the central atom(s). This describes the ligands and their mode of attachment. The κ construction can be omitted in very simple cases (see Examples 1–3 below) or when the distribution of the ligands on the central atoms is obvious (see Example 4 below).

The central atom names are listed after the ligand names. The multiplicative prefix ‘di’ is used where the central atoms are the same element. Otherwise, the order of the central atom names is obtained using Table VI. The order of the central atom names is reflected in the numbering employed with the κ symbols. The ending ‘ate’ is added if the dinuclear compound is an anion, and a radical dot may be added for radicals. In the case of two different central atoms, the two names are cited inside parentheses and ‘ate’ is added outside the parentheses.

Examples:

1. ClClO oxido-1 κO -dichlorine($\text{Cl}—\text{Cl}$),
or oxidodichlorine($\text{Cl}—\text{Cl}$)
2. $\text{ClOO}^\bullet$ chlorido-1 κCl -dioxygen($\text{O}—\text{O}$)($\bullet$),
or chloridodioxygen($\text{O}—\text{O}$)($\bullet$)
3. ClClF^+ fluoro-1 κF -dichlorine($\text{Cl}—\text{Cl}$)(1+),
or fluoridodichlorine($\text{Cl}—\text{Cl}$)(1+)
4. $[\text{O}_3\text{POSO}_3]^{2-}$ μ -oxido-hexaoxido-1 $\kappa^3 O, 2\kappa^3 O$ -(phosphorussulfur)ate(2–),
or μ -oxido-hexaoxido(phosphorussulfur)ate(2–)

5.

ethyl-2 κC -tetramethyl-1 $\kappa^3 C, 2\kappa C$ - μ -thiophene-2,5-diyl-tinbismuth

6. $[\text{Cl}(\text{PhNH})_2\text{Ge}^2\text{Ge}^1\text{Cl}_3]$ tetrachlorido-1 $\kappa^3 \text{Cl}, 2\kappa \text{Cl}$ -bis(phenylamido-2 κN)-
digermanium($\text{Ge}—\text{Ge}$)
7. $\text{Li}^1\text{Pb}^2\text{Ph}_3$ triphenyl-2 $\kappa^3 C$ -lithiumlead($\text{Li}—\text{Pb}$)

Where the precise positions of ligation are unknown, the kappa convention cannot be used.

Examples:

8. $[\text{Pb}_2(\text{CH}_2\text{Ph})_2\text{F}_4]$ dibenzyltetrafluoridodilead
9. $[\text{Ge}_2(\text{CH}_2\text{Ph})\text{Cl}_3(\text{NHPh})_2]$ (benzyl)trichloridobis(phenylamido)digermanium

IR-7.3.3

Oligonuclear compounds

In simple cases, the principles of the preceding sections may be generalized for the naming of oligonuclear compounds. Again, there are compounds which are also easily named by substitutive nomenclature because of the availability of obvious parent hydrides.

Examples:

1. $\text{HO}_3^\bullet$ hydridotrioxxygen($\bullet$)
2. $\text{HON}_3^{\bullet-}$ hydroxido-1 κO -trinitrate(2 $N—N$)($\bullet 1-$)
3. $\text{Cl}_3\text{SiSiCl}_2\text{SiCl}_3$ octachloridotrisilicon(2 $\text{Si}—\text{Si}$) (additive),
or octachlorotrisilane (substitutive)
4. $\text{FMe}_2\text{SiSiMe}_2\text{SiMe}_3$ fluoro-1 κF -heptamethyltrisilicon(2 $\text{Si}—\text{Si}$) (additive),
or 1-fluoro-1,1,2,2,3,3,3-heptamethyltrisilane (substitutive)

(An alternative additive name for the compound in Example 3, based on the longest chain in the molecule, can also be constructed by the method described in Section IR-7.4.3; see Example 6 therein.)

For heterooligonuclear systems, more conventions are needed to identify and name the collection of central atoms, and to number the central atoms so as to provide locants for the ligands.

Example:

5. $\text{Me}_3\text{SiSeSiMe}_3$
 μ -selenido-bis(trimethylsilicon) (additive), or
 hexamethyl-1 κ^3 C,2 κ^3 C-disiliconselenium(2 *Si—Se*) (additive), or
 1,1,1,3,3,3-hexamethyldisilaselenane (substitutive)

Note that in the last example one can choose to name the compound as dinuclear or trinuclear. The complexities deriving from the structural variations which may occur with homonuclear and heteronuclear central atom clusters and bridging groups are dealt with in more detail in Sections IR-9.2.5.6 to IR-9.2.5.7.

IR-7.4 INORGANIC CHAINS AND RINGS

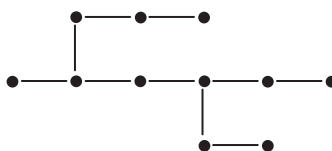
IR-7.4.1 General

Inorganic chain and ring structures may be named with no implications about the nature of bonds, except for the connectivity of the molecule or ion, using a particular system of additive nomenclature. The method can be applied to all chain and ring compounds although it is principally intended for species composed mainly of atoms other than carbon. While small molecules can be named more conveniently by using several alternative methods, the advantage of this nomenclature system lies in the simplicity with which complicated structures can be derived from the name and *vice versa*. Details of this system are given in Ref. 2; a simplified treatment is provided here.

The overall topology of the structure is specified as follows. A neutral chain compound is called 'catena' preceded by a multiplicative prefix, 'di', 'tri', *etc.*, to indicate the number of branches in the molecule. Likewise, cyclic compounds are called 'cycle' preceded by the appropriate multiplicative prefix. A mixed chain and ring compound is classified as an *assembly* composed of acyclic and cyclic *modules* and, if neutral, is named as 'catenacycle', with appropriate multiplicative prefixes inserted as in Example 3 below.

Examples:

1.



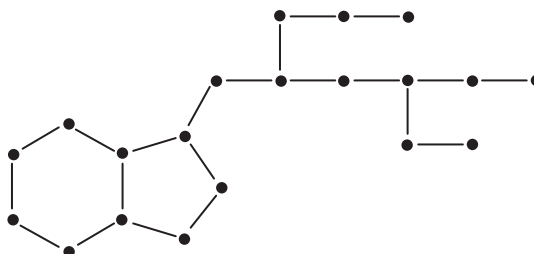
tricatena

2.



dicycle

3.



tricenadicycle

IR-7.4.2

Nodal descriptor

The connectivity in the molecular framework is indicated by a *nodal descriptor*, which is placed in square brackets immediately before the terms ‘catena’, ‘cycle’ or ‘catenacycle’. The atoms are numbered according to the general nodal nomenclature regardless of their identity. Only in the case of ambiguity are the identities of the atoms taken into consideration.

The first part of the descriptor indicates the number of atoms in the main chain. The arabic numerals after the full stop indicate the lengths of the branches cited in priority order. A superscript locant for each branch denotes the atom in the part of the molecule already numbered to which the branch is attached.

A zero in the descriptor indicates a ring and is followed by an arabic numeral indicating the number of atoms in the main ring. For polycyclic systems, the numbering begins from one of the bridgeheads and proceeds in the direction which gives the lowest possible locant for the other bridgehead. In this case, the number of atoms in the bridge is cited after the full stop. A pair of superscript locants is inserted for each such bridge numeral, separated by a comma and cited in increasing numerical order.

An assembly descriptor consists of square brackets enclosing, in the order of their seniority (see Ref. 2 for rules), the nodal descriptors of each module in parentheses. Between the descriptors of the modules, the locants of the nodes linking the modules are indicated. These locants, separated by a colon, are the atom numbers in the *final* numbering of the entire assembly (compare Example 7 below with Examples 5 and 6).

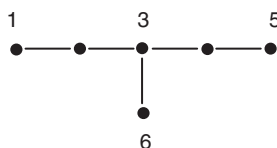
Examples:

1.



descriptor: [7]

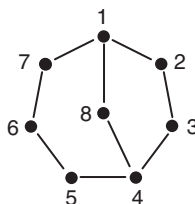
2.

descriptor: $[5.1^3]$

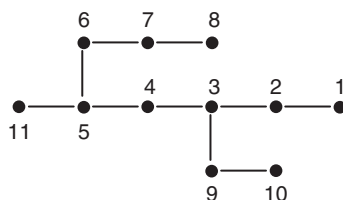
3.

descriptor: $[06]$

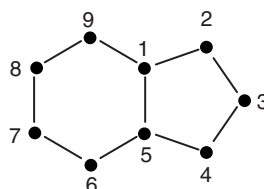
4.

descriptor: $[07.1^{1,4}]$

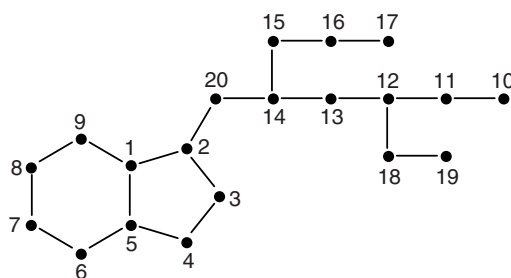
5.

descriptor: $[8.2^3 1^5]$

6.

descriptor: $[09.0^{1,5}]$

7.

descriptor: $[(09.0^{1,5})2:20(8.2^3 1^5)]$

IR-7.4.3

Name construction

The atoms forming the nodal skeleton are listed in alphabetical order complete with their locants and are named using 'y' terms, examples of which are given in Table IR-7.1; a full list is given in Table X.

Table IR-7.1 Some 'y' terms for naming elements in the nodal framework

H	hydrony	C	carby	N	azy	O	oxy
B	bory	Si	sily	P	phosphy	S	sulfy
		Ge	germy	As	arsy	Se	seleny
		Sn	stanny	Sb	stiby	Te	tellury

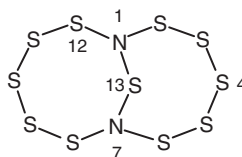
Atoms and groups of atoms which are not part of the nodal framework are named as ligands (Section IR-7.1.3) and are cited in alphabetical order, together with their locants, before the sequence of names of the atoms constituting the nodal framework. The nodal descriptor is given next. The 'catena', 'cycle' or 'catenacycle' term is added at the end, *cf.* Section IR-7.4.1. (Note that bridging ligands are not employed in this system.)

In the case of anionic and cationic species, these terms are modified by the endings 'ate' and 'ium' respectively, to yield the terms 'catenate', 'catenium', 'cyclate', 'cyclium', 'catenadicyclium', 'catenacyclate', *etc.*, and a charge number is added at the end of the name. Radical species may be indicated analogously by using the radical dot (see Section IR-7.1.4).

Examples 1–6, which demonstrate the use of the system described here, were also named in Section IR-7.3.1. Examples 7–13 cannot be named so easily by other methods.

Examples:

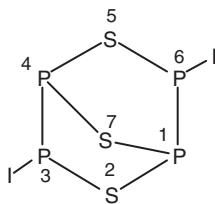
- | | |
|--|---|
| 1. NCCN | 1,4-diazy-2,3-dicarby-[4]catena |
| 2. NCCN ^{•-} | 1,4-diazy-2,3-dicarby-[4]catenate(•1-) |
| 3. NCSSCN | 1,6-diazy-2,5-dicarby-3,4-disulfy-[6]catena |
| 4. NCSSCN ^{•-} | 1,6-diazy-2,5-dicarby-3,4-disulfy-[6]catenate(•1-) |
| 5. HSSH ^{•-} | 1,2-dihydrido-1,2-disulfy-[2]catenate(•1-) |
| 6. Cl ₃ SiSiCl ₂ SiCl ₃ | 2,2,3,3,4,4-hexachlorido-1,5-dichlory-2,3,4-trisily-[5]catena |
| 7. ClSiH ₂ SiH(Me)NSO | 2,2,3-trihydrido-3-methyl-4-azy-1-chlory-6-oxy-2,3-disily-5-sulfy-[6]catena |
| 8. | |



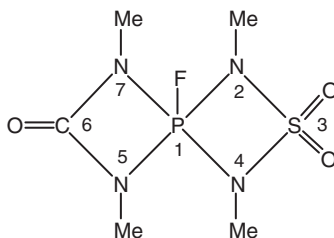
1,7-diazyundecasulfy-[012.1^{1,7}]dicycle

Because this compound contains only nitrogen and sulfur it is not necessary to indicate the locants of all sulfur atoms. Only the locants of the two nitrogen atoms are needed. The same applies to several of the following examples.

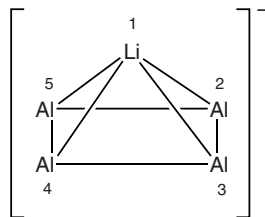
9.

3,6-diiodido-1,3,4,6-tetraphosphy-2,5,7-trisulphy-[06.1^{1,4}]dicycle

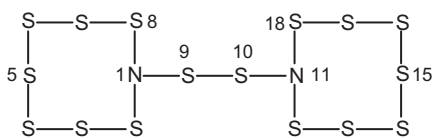
10.

1-fluorido-2,4,5,7-tetramethyl-3,3,6-trioxido-2,4,5,7-tetraazy-6-carby-1-phosphy-3-sulphy-[04.3^{1,1}]dicycle

11.

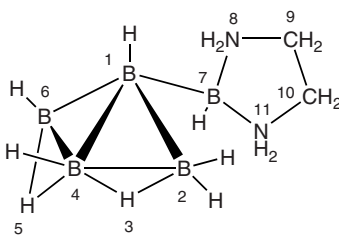
tetraaluminy-1-lithy-[05.0^{1,30^1,40^2,5}]tetracyclate(1⁻)

12.



1,11-diazyhexadecasulphy-[(08)1:9(2)10:11(08)]catenadicycle

13.

1,2,2,4,6,7,8,8,9,9,10,10,11,11-tetradecahydrido-8,11-diazy-1,2,4,6,7-pentaboro-9,10-dicarby-3,5-dihydrony-[(06.0^{1,40^2,40^4,6})1:7(05)]pentacycle

IR-7.5

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